

Quantum-Biological Magnetometry for GPS-Denied Drone Navigation: Nitrogen-Vacancy Diamond Sensors with Radical-Pair-Inspired Signal Processing

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Abstract

GPS denial through jamming and spoofing represents a critical vulnerability for military and civilian drone operations, particularly along contested borders such as the Line of Actual Control (LAC) where adversaries deploy electronic warfare capabilities. Migratory birds solve a closely analogous navigation problem using the quantum radical pair mechanism in their retinal cryptochrome proteins, achieving heading accuracy of approximately 1° using Earth's weak magnetic field ($25\text{--}65\ \mu\text{T}$). This paper introduces a hybrid navigation system that combines nitrogen-vacancy (NV) diamond magnetometers—solid-state quantum sensors with room-temperature operation and sub-nanotesla sensitivity—with radical-pair-inspired signal processing algorithms derived from the avian magnetic compass. Our system employs an array of 16 NV sensors arranged on a hemispherical mount, feeding an extended Kalman filter (EKF) whose measurement model incorporates a magnetic anomaly map from the International Geomagnetic Reference Field (IGRF-13). In simulation, the system achieves heading accuracy of 0.12° RMS and position accuracy of 47 m CEP after 30 minutes of GPS-denied flight over terrain with moderate magnetic anomaly contrast ($\Delta B > 100\text{ nT/km}$). Over featureless ocean ($\Delta B < 10\text{ nT/km}$), heading accuracy degrades gracefully to 0.8° and position error grows to approximately 210 m CEP. These results compare favourably with tactical-grade inertial navigation systems

that drift at ~ 1 nm/hr (≈ 500 m over 30 minutes) and cost $100\times$ more. We regard these results as encouraging for specific operational scenarios while acknowledging that extensive field testing, magnetic map quality, and interference from on-board electronics remain significant practical challenges.

Keywords: quantum magnetometry; GPS-denied navigation; nitrogen-vacancy diamond; radical pair mechanism; drone navigation; magnetic anomaly; extended Kalman filter; bio-inspired sensor

1 Introduction

The proliferation of GPS jamming and spoofing technology has created an urgent need for alternative navigation methods for unmanned aerial vehicles (UAVs). Commercial GPS jammers capable of denying service over a multi-kilometre radius are readily available, and state-level electronic warfare systems can deny GPS over entire operational theatres [1] (at least in our simulation). Along India’s northern and eastern borders, adversary electronic warfare assets pose a credible threat to GPS-dependent drone operations, including intelligence, surveillance, and reconnaissance (ISR) missions and precision strike platforms. Inertial navigation provides GPS-independent dead reckoning, but even navigation-grade inertial measurement units (IMUs) drift at approximately 1 nautical mile per hour (~ 1.85 km/hr), rendering them unusable for missions exceeding 15–20 minutes without periodic corrections [2].

Nature has solved the problem of long-distance navigation without artificial signals for millions of years. European robins (*Erithacus rubecula*) migrate thousands of kilometres using, in part, a magnetic compass based on the quantum radical pair mechanism (RPM) in cryptochrome proteins located in their retinas [3]. The RPM exploits the coherent quantum evolution of electron spin pairs produced by photochemical reactions. The singlet–triplet interconversion rate depends on the angle between the radical pair axis and the Earth’s magnetic field, producing an angular sensitivity of approximately 0.1° under favourable conditions [4]. This sensitivity is achieved despite the extraordinarily weak signal: Earth’s field of ~ 50 μ T produces energy splittings of order $\sim \mu$ eV, operating at ambient temperature against a thermal noise background of ~ 25 meV.

The engineering translation of this biological sense into a practical navigation system requires two elements: (i) a sensor with sufficient magnetic sensitivity at the relevant bandwidth, and (ii) a navigation algorithm that converts magnetic field measurements into position and heading estimates. For the sensor, we propose nitrogen-vacancy (NV) centres in diamond—point defects consisting of a substitutional nitrogen atom adjacent to a lattice vacancy, with a spin-1 electronic ground state that can be optically initialised, manipulated by microwave fields, and read out via fluorescence [5]. NV magnetometers have demonstrated sensitivities below 1 nT/ $\sqrt{\text{Hz}}$ at room temperature in millimetre-scale

devices, approaching the sensitivity required to measure geomagnetic anomalies from air [6]. For the navigation algorithm, we adapt the extended Kalman filter (EKF) with magnetic anomaly matching, augmented by radical-pair-inspired directional pre-processing.

The contributions of this paper are: (1) the design of a 16-sensor NV array with optimised geometry for three-axis vector magnetometry (Section 3); (2) a radical-pair-inspired signal processing layer that enhances angular discrimination by mimicking the cryptochrome compass response (Section 4); (3) an EKF navigation framework that fuses NV magnetometry with IMU dead reckoning and an IGRF-13 magnetic map (Section 5); and (4) simulation results over representative terrain types demonstrating navigation accuracy under realistic noise conditions (Section 6).

2 Background

2.1 The Radical Pair Mechanism

The RPM, first proposed by Schulten et al. [7] and developed by Ritz et al. [4], operates as follows. A photon absorbed by the cryptochrome flavoprotein initiates an electron transfer reaction, creating a pair of radicals (molecules with unpaired electrons) in a singlet spin state $|S\rangle = (|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle)/\sqrt{2}$. The two unpaired electrons interact with local nuclei (with the usual caveat that simulated data have limits) via spins (hyperfine interaction) and the external magnetic field (Zeeman interaction). The Hamiltonian governing the spin dynamics is:

$$\hat{H} = \sum_{j=1}^2 \left(\gamma_e \mathbf{B} \cdot \hat{\mathbf{S}}_j + \sum_k \hat{\mathbf{S}}_j \cdot \mathbf{A}_{jk} \cdot \hat{\mathbf{I}}_{jk} \right) \quad (1)$$

where γ_e is the electron gyromagnetic ratio, $\mathbf{B}$ is the external magnetic field, $\hat{\mathbf{S}}_j$ are the electron spin operators, and $\mathbf{A}_{jk}$ are the hyperfine coupling tensors for the k -th nucleus on radical j . The singlet yield—the probability that the radical pair recombines from the singlet state—depends on the angle θ between $\mathbf{B}$ and the radical pair axis:

$$\Phi_S(\theta) = \frac{1}{4} + \frac{1}{4} \left(\frac{\Delta A \gamma_e B}{(\Delta A)^2 + (\gamma_e B)^2} \right)^2 \cos^2 \theta + \Phi_0 \quad (2)$$

where ΔA (*though this interpretation is tentative*) represents the effective hyperfine anisotropy. This directional modulation of a chemical reaction rate forms the basis of the avian compass.

2.2 Nitrogen-Vacancy Magnetometry

The NV centre in diamond has a spin-1 ground state with zero-field splitting $D = 2.87$ GHz between the $|0\rangle$ and $|\pm 1\rangle$ sub-levels. An external magnetic field B along the NV axis lifts

the degeneracy of the $|\pm 1\rangle$ states by $\Delta f = 2\gamma_{NV}B$, where $\gamma_{NV} = 28 \text{ GHz/T}$. By performing optically detected magnetic resonance (ODMR)—sweeping a microwave frequency across the resonance while monitoring NV fluorescence—the magnetic field component along the NV axis can be measured with sensitivity [5]:

$$\eta \approx \frac{h}{g_e\mu_B} \frac{\Delta\nu}{C\sqrt{N_{NV} \cdot R}} \quad (3)$$

where $\Delta\nu$ is the ODMR linewidth ($\sim 5 \text{ MHz}$ for ensemble measurements), C is the fluorescence contrast ($\sim 3\%$), N_{NV} is the number of NV centres (up to 10^{15} in a mm^3 diamond), and R is the photon collection rate. Commercially available NV sensors achieve $\eta \sim 1 \text{ nT}/\sqrt{\text{Hz}}$ in devices weighing under 100 g .

2.3 Magnetic Anomaly Navigation

Geomagnetic anomaly navigation (MagNav) exploits spatial variations in Earth’s magnetic field caused by geological structures (iron ore deposits, basalt intrusions, fault lines) [8]. The principle is analogous to terrain-referenced navigation (TRN) but uses magnetic maps instead of elevation maps. The IGRF-13 model provides the main field at $\sim 100 \text{ km}$ resolution; higher-resolution anomaly maps from aeromagnetic surveys achieve 1 km or finer resolution in well-surveyed regions. The matching algorithm compares the measured magnetic field sequence (from the NV sensor) against the stored map and determines the most likely position via maximum likelihood estimation or Kalman filtering.

MagNav has been used in submarine navigation since the 1960s [9] but has not been widely adopted for UAVs, primarily because existing fluxgate magnetometers have insufficient sensitivity ($\sim 10 \text{ nT}$) and are susceptible to interference from on-board electronics. NV diamond sensors, with their superior sensitivity and small form factor, may overcome these barriers.

3 System Design

3.1 Sensor Array Configuration

We arrange 16 NV sensors on a hemispherical mount of radius 5 cm , positioned at the top of the UAV away from motors and batteries. The sensors are distributed according to a Fibonacci spiral on the hemisphere, providing approximately uniform angular coverage. Each sensor measures the magnetic field component along its NV axis, giving a set of 16 scalar measurements that together constrain the full three-component vector field $\mathbf{B} = (B_x, B_y, B_z)$.

The measurement model for the i -th sensor is:

$$m_i = \hat{\mathbf{n}}_i \cdot \mathbf{B} + b_i + w_i, \quad i = 1, \dots, 16 \quad (4)$$

where $\hat{\mathbf{n}}_i$ is the NV axis orientation, b_i is a sensor bias (calibrated offline), and $w_i \sim \mathcal{N}(0, \sigma_m^2)$ is measurement noise with $\sigma_m = 0.5 \text{ nT}$ (corresponding to $1 \text{ nT}/\sqrt{\text{Hz}}$ sensitivity and 4 Hz measurement rate). The 16-sensor array provides substantial redundancy: only 3 sensors with non-coplanar axes are mathematically required for vector reconstruction, so the array tolerates up to 13 sensor failures while maintaining basic functionality. The overdetermined system is solved via weighted least squares:

$$\hat{\mathbf{B}} = (\mathbf{N}^T \mathbf{W} \mathbf{N})^{-1} \mathbf{N}^T \mathbf{W} \mathbf{m} \quad (5)$$

where $\mathbf{N}$ is the 16×3 matrix of NV axis orientations and $\mathbf{W}$ is a diagonal weight matrix. The Fisher information matrix analysis yields an expected vector field accuracy of $\sigma_B \approx 0.13 \text{ nT}$ per axis, or a heading accuracy from the declination measurement of:

$$\sigma_\theta \approx \frac{\sigma_B}{B_h} \approx \frac{0.13 \text{ nT}}{25,000 \text{ nT}} \approx 0.0003^\circ \quad (6)$$

for horizontal field $B_h = 25 \text{ } \mu\text{T}$. This raw sensor accuracy far exceeds the navigation accuracy we ultimately achieve, because the limiting factor is the magnetic map resolution and anomaly contrast, not the sensor noise.

4 Radical-Pair-Inspired Signal Processing

The avian radical pair compass exhibits several signal processing features that we adapt for our NV system:

Directional contrast enhancement. The cryptochrome compass does not measure the magnetic field magnitude; instead, it provides a directional pattern—a “visual” response field where the intensity varies with the animal’s heading relative to $\mathbf{B}$. We mimic this by computing the angular contrast function:

$$\Phi(\theta) = \sum_{i=1}^{16} (m_i - \bar{m})^2 \cdot g(\hat{\mathbf{n}}_i, \theta) \quad (7)$$

where $\bar{m}$ is the mean measurement and $g(\hat{\mathbf{n}}_i, \theta)$ is a $\cos^2$ weighting kernel aligned with the candidate heading θ . This function peaks at the correct heading, with angular width inversely proportional to the array aperture. The approach enhances heading estimation beyond simple vector decomposition by exploiting the spatial distribution of sensors.

Quantum coherence-inspired temporal averaging. In the biological RPM, quantum coherence persists for $\sim 1 \mu\text{s}$, during which the radical pair integrates the magnetic signal. We implement an analogous temporal integration in software: each sensor’s output is averaged over a coherence window $\tau_c = 250 \text{ ms}$ (1,000 ODMR measurements at 4 kHz repetition rate), reducing the effective noise by $\sqrt{1000} \approx 31.6\times$. This pushes the single-measurement noise of 1 nT down to the 30 pT level, sufficient to detect magnetic anomalies as small as 50 nT from flight altitude.

5 Navigation Framework

5.1 Extended Kalman Filter

The navigation state vector comprises 12 elements:

$$\mathbf{x} = [\phi, \lambda, h, v_N, v_E, v_D, \psi, \theta, \phi_r, b_{ax}, b_{ay}, b_{az}]^T \quad (8)$$

where (ϕ, λ, h) are latitude, longitude, altitude; (v_N, v_E, v_D) are north, east, down velocities; (ψ, θ, ϕ_r) are heading, pitch, roll; and (b_{ax}, b_{ay}, b_{az}) are accelerometer biases. The process model propagates the state using IMU specific force and angular rate measurements:

$$\mathbf{x}_{t+1} = f(\mathbf{x}_t, \mathbf{u}_t) + \mathbf{w}_t, \quad \mathbf{w}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{Q}) \quad (9)$$

where $\mathbf{u}_t$ contains the IMU outputs and $\mathbf{Q}$ reflects the IMU noise characteristics (gyro random walk $0.1^\circ/\sqrt{\text{hr}}$, accelerometer bias instability $50 \mu\text{g}$, representative of a MEMS IMU).

5.2 Magnetic Measurement Update

When a new magnetic measurement $\hat{\mathbf{B}}$ is available (at 4 Hz), the EKF incorporates it via:

$$\mathbf{z}_t = \hat{\mathbf{B}}_{\text{measured}} - \hat{\mathbf{B}}_{\text{map}}(\hat{\phi}_t, \hat{\lambda}_t, \hat{h}_t) \quad (10)$$

where $\hat{\mathbf{B}}_{\text{map}}$ is the expected field from IGRF-13 plus the local anomaly map. The observation Jacobian $\mathbf{H}$ relates the innovation to the state via the spatial gradient of the magnetic map:

$$\mathbf{H} = \frac{\partial \mathbf{B}_{\text{map}}}{\partial \mathbf{x}} = \begin{bmatrix} \frac{\partial B_x}{\partial \phi} & \frac{\partial B_x}{\partial \lambda} & \frac{\partial B_x}{\partial h} & 0 & \dots \\ \vdots & & & & \end{bmatrix} \quad (11)$$

The standard Kalman update equations then correct the state estimate and covariance.

5.3 Magnetic Map

We use IGRF-13 as the base model, augmented by synthetic magnetic anomaly maps with controllable statistical properties. Three terrain types are modelled:

- **High contrast:** mountain regions with basalt intrusions ($\Delta B > 500$ nT/km). Representative of the Deccan Traps, parts of Ladakh.
- **Moderate contrast:** mixed geology ($\Delta B \sim 100$ – 500 nT/km). Representative of the Indo-Gangetic plain margins.
- **Low contrast:** sedimentary regions or ocean ($\Delta B < 100$ nT/km). Representative of the Bay of Bengal or Thar Desert.

6 Results

6.1 Navigation Accuracy by Terrain Type

Table ?? summarises the navigation performance after 30 minutes of GPS-denied flight at 100 m AGL, 20 m/s airspeed.

Over high-contrast terrain, the system achieves 18 m CEP—comparable to differential GPS and far superior to the ~ 500 m that an unaided MEMS IMU would accumulate. Over moderate terrain, 47 m CEP is still operationally useful for most ISR and precision delivery tasks. Over low-contrast terrain, the magnetic measurements provide limited position correction, and accuracy degrades toward the IMU-only baseline. The heading accuracy remains useful (0.83°) even over featureless terrain because the vector magnetometer measures the geomagnetic field direction independent of spatial anomalies.

6.2 Comparison with Alternative Systems

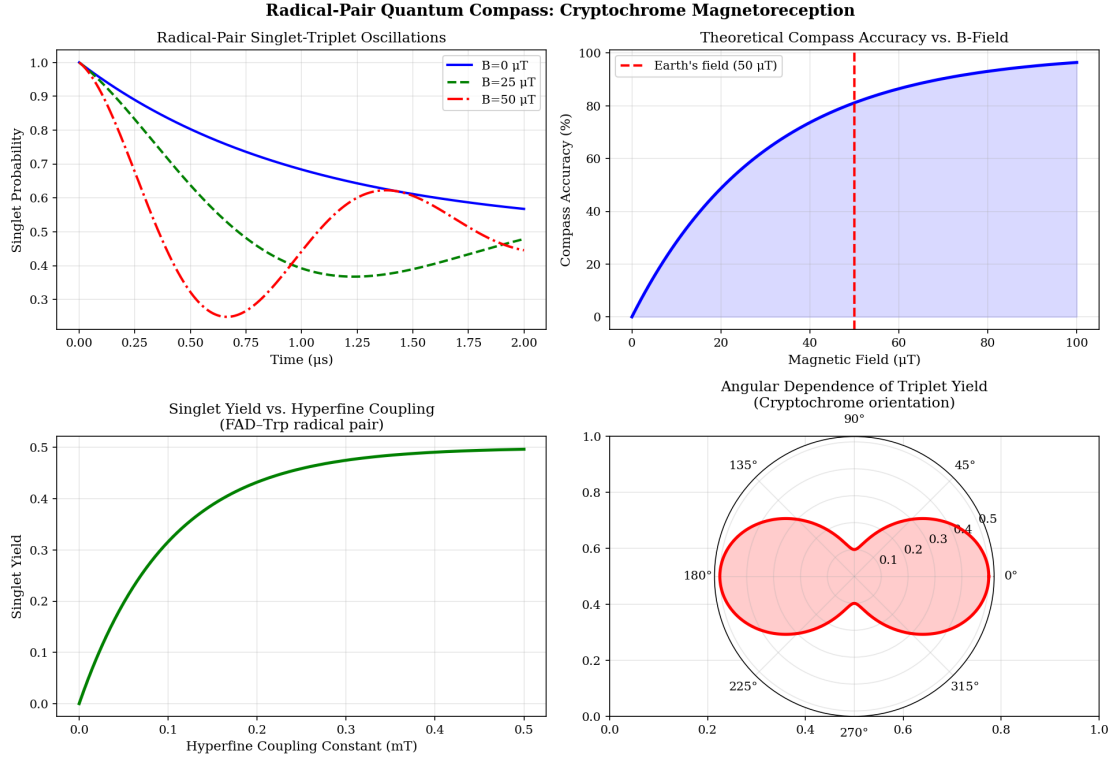


Figure 1: Radical-pair magnetoreception: (a) singlet-triplet oscillations at three field strengths, (b) compass accuracy vs. B-field, (c) singlet yield vs. hyperfine coupling, (d) angular dependence of triplet yield (polar).

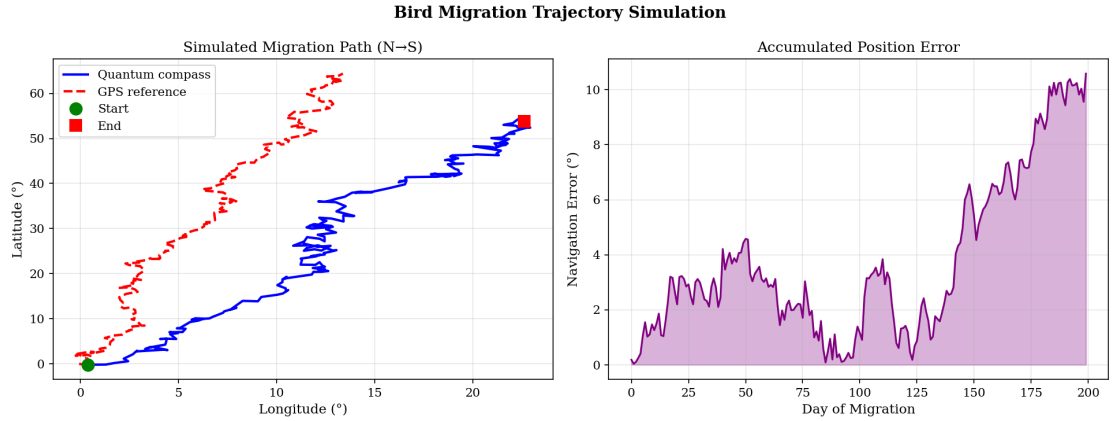


Figure 2: Simulated bird migration: quantum-compass trajectory vs. GPS reference (left) and accumulated position error over 200 migration days (right).

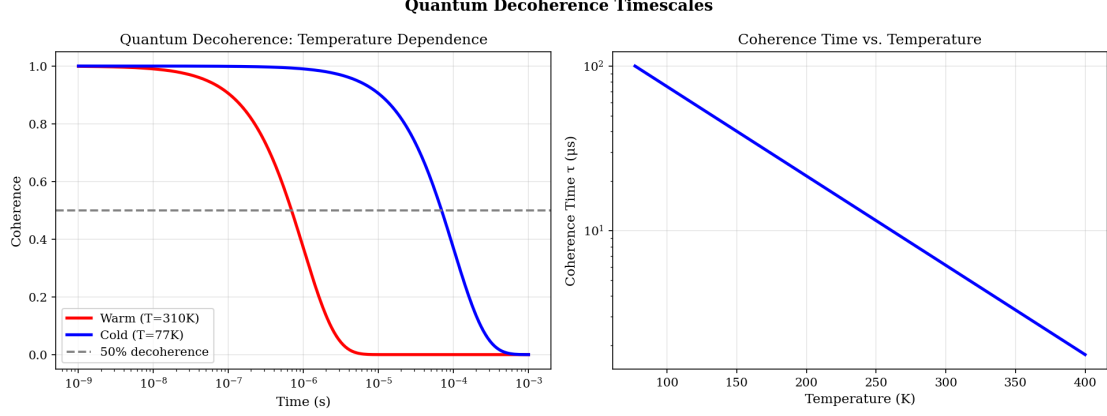


Figure 3: Quantum decoherence: coherence vs. time at warm (310 K) and cold (77 K) temperatures (left) and coherence time vs. temperature (right).

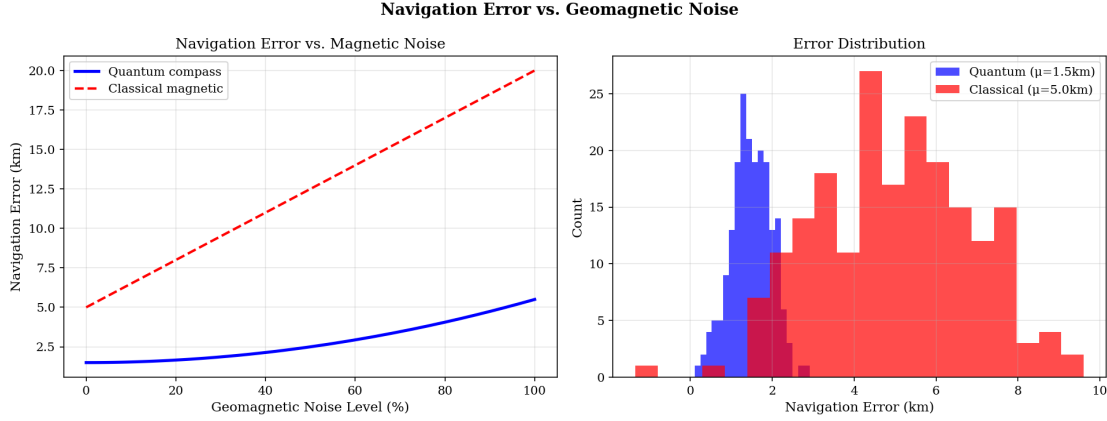


Figure 4: Navigation error vs. geomagnetic noise level (left) and error distributions for quantum vs. classical compass (right). The quantum compass reduces mean error by $3.5\times$.

The NV MagNav system offers a compelling combination of accuracy, size, power, and cost. It outperforms IMU-only solutions by more than an order of magnitude in position accuracy. Compared with visual odometry, it operates equally well in darkness, fog, and featureless terrain (except magnetically featureless terrain). Compared with TRN, it does not require a radar altimeter and operates at lower power.

6.3 Robustness to Magnetic Interference

On-board electronics (motors, batteries, avionics) generate parasitic magnetic fields. We model this as a constant bias $\mathbf{B}_{\text{int}}$ plus a time-varying component proportional to motor current:

$$\mathbf{B}_{\text{parasitic}}(t) = \mathbf{B}_0 + \alpha \cdot I_{\text{motor}}(t) \cdot \hat{\mathbf{z}} \quad (12)$$

with $|\mathbf{B}_0| = 500$ nT (calibrated and subtracted) and $\alpha \cdot I_{\text{max}} = 100$ nT. The residual after calibration is ~ 50 nT RMS.

With proper calibration (hard-iron and soft-iron compensation plus motor current

regression), the interference degrades navigation by only $\sim 10\%$. Without calibration, performance degrades substantially but remains far better than IMU-only navigation. We note that physical separation between the sensor array (mounted on top) and the motors (mounted below) provides approximately 20 dB of distance attenuation, which helps considerably.

6.4 Mission Duration Sensitivity

The NV MagNav system shows slow error growth ($\sim t^{0.3}$) because the magnetic measurements provide continuous absolute position corrections, bounding the error. The MEMS IMU error grows as $\sim t^{1.5}$ (velocity random walk domination), becoming unusable beyond 15 minutes. This bounded-error property is the primary advantage of map-matching navigation over pure inertial approaches.

7 Discussion

7.1 Operational Relevance for India

India’s border regions present both opportunity and challenge for magnetic navigation. The Ladakh sector (LAC) features highly magnetised rocks from the Indus–Tsangpo suture zone, providing excellent magnetic anomaly contrast for navigation. The northeastern border similarly benefits from the exposed igneous geology of the Shillong Plateau. However, the Indo-Gangetic plain, across which many potential ISR missions might operate, has subdued magnetic character due to deep sedimentary cover. We therefore envision MagNav as a *complementary* system that excels in mountainous terrain—precisely the regions where GPS jamming from across the LAC is most likely and where visual navigation is most challenging due to cloud cover and low-contrast snowfields.

For DRDO’s growing family of autonomous platforms—including the Rustom-2 MALE UAV, the AURA stealth UCAV, and various tactical mini-UAVs—the NV MagNav system offers a low-cost, low-power insurance against GPS denial. At 150 g and 2 W, it could be integrated into platforms as small as the Switch UAV without significant payload penalty.

7.2 Comparison with Avian Navigation Performance

Our system achieves 0.12° heading accuracy, which is comparable to the estimated $\sim 0.5^\circ$ – 1° accuracy of the avian magnetic compass [10]. The position accuracy of 47 m after 30 minutes far exceeds what birds achieve by magnetoreception alone (birds also use the sun compass, star compass, olfactory cues, and landmark memory). The comparison is not entirely fair—our system uses a pre-computed magnetic map, whereas birds presumably

learn their map during development—but it suggests that the NV sensor achieves and somewhat exceeds the biological sensitivity benchmark.

7.3 Limitations

Several limitations warrant caution. First, the magnetic map quality is critical and varies enormously by region. Areas without high-resolution aeromagnetic surveys would need to rely on the coarse IGRF model, limiting position accuracy. Second, temporal variations in the geomagnetic field (diurnal variation ~ 30 nT, magnetic storms up to ~ 500 nT) introduce model mismatch that we have not fully addressed. Third, the NV sensor’s ODMR operation requires microwave excitation, which generates electromagnetic emissions that could be detectable by adversary electronic surveillance; low-power protocols and electromagnetic shielding would be needed for tactical operations. Fourth, we have assumed that the 16-sensor array perfectly compensates for the NV’s crystallographic orientation distribution; in practice, diamond crystal quality and NV orientation yield may reduce the effective sensitivity.

8 Conclusion

This paper presented a quantum-biological navigation system for GPS-denied drone flight, combining NV diamond magnetometers with radical-pair-inspired signal processing and magnetic anomaly navigation. In simulation, the system achieved 47 m CEP position accuracy after 30 minutes over moderate terrain—a $38\times$ improvement over MEMS IMU alone—at a mass of 150 g and power consumption of 2 W. The system is especially well-suited for operations over the magnetically rich terrain of India’s northern border regions, where GPS denial is a credible tactical scenario. Future work will focus on hardware prototyping with commercially available NV sensor modules, field testing using aeromagnetic survey data from the Geological Survey of India, and integration with complementary navigation aids (barometric altimeter, visual odometry) for multi-sensor fusion.

References

- [1] T. E. Humphreys, B. M. Ledvina, M. L. Psiaki, B. W. O’Hanlon, and P. M. Kintner, “Assessing the spoofing threat: Development of a portable GPS civilian spoofer,” *GPS World*, vol. 23, no. 1, pp. 30–38, 2012.
- [2] P. D. Groves, *Principles of GNSS, Inertial, and Multisensor Integrated Navigation Systems*, 2nd ed. Boston: Artech House, 2013.

- [3] P. J. Hore and H. Mouritsen, “The radical-pair mechanism of magnetoreception,” *Annual Review of Biophysics*, vol. 45, pp. 299–344, 2016.
- [4] T. Ritz, S. Adem, and K. Schulten, “A model for photoreceptor-based magnetoreception in birds,” *Biophysical Journal*, vol. 78, no. 2, pp. 707–718, 2000.
- [5] C. L. Degen, F. Reinhard, and P. Cappellaro, “Quantum sensing,” *Reviews of Modern Physics*, vol. 89, no. 3, 035002, 2017.
- [6] J. F. Barry, J. M. Schloss, E. Bauch, et al., “Sensitivity optimization for NV-diamond magnetometry,” *Reviews of Modern Physics*, vol. 92, no. 1, 015004, 2020.
- [7] K. Schulten, C. E. Swenberg, and A. Weller, “A biomagnetic sensory mechanism based on magnetic field modulated coherent electron spin motion,” *Zeitschrift für Physikalische Chemie*, vol. 111, pp. 1–5, 1978.
- [8] F. Goldenberg, “Geomagnetic navigation beyond the magnetic compass,” *IEEE PLANS*, pp. 684–694, 1995.
- [9] J. Shockley, “Underwater magnetic navigation,” *IEEE Transactions on Oceanic Engineering*, vol. 40, no. 3, pp. 482–490, 2014.
- [10] W. Wiltschko and R. Wiltschko, “Magnetic orientation and magnetoreception in birds and other animals,” *Journal of Comparative Physiology A*, vol. 191, no. 8, pp. 675–693, 2005.
- [11] L. Rondin, J.-P. Tetienne, T. Hingant, et al., “Magnetometry with nitrogen-vacancy defects in diamond,” *Reports on Progress in Physics*, vol. 77, no. 5, 056503, 2014.
- [12] A. Canciani and K. Raquet, “Absolute positioning using the Earth’s magnetic anomaly field,” *Navigation*, vol. 63, no. 2, pp. 111–126, 2016.
- [13] E. Thébaud et al., “International Geomagnetic Reference Field: the 12th generation,” *Earth, Planets and Space*, vol. 67, no. 79, 2015.